Preprint manuscript No.
(will be inserted by the editor)

The Spectral Entropy-Fold (SEF) Hypothesis: A Geometric Framework Connecting Discrete Ricci Curvature, Spectral Graph Theory, and NP-Complexity

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June 26, 2026

Abstract We formulate the Spectral Entropy-Fold (SEF) hypothesis as a geometric research program for studying whether structural hardness in NP-complete instances can be represented through discrete curvature, spectral dispersion, and kernel embeddings. The manuscript does not claim a resolution of P versus NP. Its more limited aim is to define a testable bridge: construct a weighted problem graph, embed that graph into a reproducing-kernel Hilbert space, measure curvature and spectral regularity on the induced manifold, and determine whether a restricted instance class thereby acquires optimization-friendly geometry. We define the Entropy-Fold operator, state a conditional Ricci-Convexity Lemma preserving a spectral-gap lower bound under explicit curvature hypotheses, distinguish proved statements from conjectural implications, and specify measurable claims, falsifiers, and validation criteria. The result is a tighter mathematical scaffold for a geometric complexity hypothesis.

Keywords Complexity theory · Discrete Ricci curvature · Spectral graph theory · RKHS embeddings · Information geometry

1 Introduction

The P versus NP problem asks whether every decision problem whose solutions can be verified in polynomial time can also be solved in polynomial time. The problem remains unresolved, and several barrier results explain why broad classes of familiar proof techniques are unlikely to suffice (Cook, 1971; Baker et al, 1975; Razborov and Rudich, 1997; Aaronson and Wigderson, 2009). One response to this impasse is to abandon direct class-separation rhetoric and instead isolate structural properties that may distinguish hard instances from tractable ones.

The present paper studies one such property. Discrete Ricci curvature on graphs and metric spaces captures how local neighborhoods expand, contract, and mix under transport (Ollivier, 2009; Lin et al, 2011). Spectral graph theory records expansion and diffusion through eigenvalue structure (Chung, 1997). Information

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geometry treats families of models as curved spaces whose geometry encodes inferential difficulty (Amari, 2016). The SEF hypothesis places these strands in a common framework by asking whether an NP-complete instance can be associated with an embedded geometric object whose curvature and spectral profile carry information about combinatorial hardness.

Scope matters here. The Ricci-Convexity Lemma proved below is conditional and internal to the proposed framework. The SEF hypothesis itself is conjectural. Most importantly, the missing bridge from spectral regularity to polynomial-time global optimization is treated as the main open step rather than as an implicit conclusion.

Contributions.

1. We formalize a weighted-problem-graph to RKHS pipeline via the Entropy-Fold operator.
2. We introduce explicit notation for spectral entropy, embedded curvature observables, and optimization-relevant regularity.
3. We state and prove a conditional Ricci-Convexity Lemma preserving a spectral-gap lower bound under a Bakry–Émery curvature-dimension assumption.
4. We convert the original SEF idea into a research-standard theory paper with proof status separation, validation logic, and falsification criteria.

2 Formal Setup

Let x be an instance of an NP-complete optimization or decision problem of size n, and let $G_x = (V_x, E_x, w_x)$ denote an associated weighted problem graph.

Definition 1 (Weighted problem graph) For instance x, the graph $G_x = (V_x, E_x, w_x)$ has candidate states V_x, local-transition edges $E_x \subseteq V_x \times V_x$, and nonnegative edge or state penalty weights w_x encoding objective or consistency costs. We assume V_x and E_x are generated by a polynomial-time local-description procedure, even if $|V_x|$ itself is exponential in n.

Definition 2 (Normalized Laplacian and spectral gap) Let A_x be the weighted adjacency matrix of G_x and D_x the diagonal degree matrix. The normalized Laplacian is

$$\mathcal{L}_x = I - D_x^{-1/2} A_x D_x^{-1/2}.$$

If $0 = \lambda_1 \leq \lambda_2 \leq \cdots \leq \lambda_m$ are its eigenvalues, then λ_2 is the spectral gap.

Definition 3 (Spectral entropy) Let $Z_x = \sum_{i:\lambda_i>0} \lambda_i$. The spectral entropy of G_x is

$$H(G_x) = -\sum_{i:\lambda_i>0} \frac{\lambda_i}{Z_x} \log\left(\frac{\lambda_i}{Z_x}\right).$$

Large $H(G_x)$ indicates diffuse nonzero spectral mass, whereas small $H(G_x)$ indicates stronger concentration.

Table 1 Core notation for the SEF framework.

Symbol	Meaning
G_x	weighted problem graph for instance x
$\mathcal{L}_x$	normalized graph Laplacian
λ_2	spectral gap of $\mathcal{L}_x$
$H(G_x)$	spectral entropy
Φ_S	Entropy-Fold embedding operator
M_x	embedded manifold in RKHS
f_x	embedded objective on M_x
$\mathsf{Curv}(x)$	event that the required curvature condition holds

Definition 4 (Entropy-Fold operator) Let $\mathcal{H}_k$ be an RKHS with Gaussian kernel

$$k(u,v) = \exp\left(-\frac{\|u-v\|_2^2}{2\sigma^2}\right).$$

The Entropy-Fold operator is a map

$$\Phi_S : G_x \longrightarrow M_x \subseteq \mathcal{H}_k$$

constructed from a finite-dimensional kernel-PCA representation of local graph descriptors. In coordinates,

$$\Phi_S(v) = \sum_{j=1}^{d} \alpha_j \, k(v, v_j) e_j,$$

where $\{v_j\}_{j=1}^d$ are anchor states, α_j are kernel-PCA coefficients, and d is chosen to retain a declared fraction $(1-\varepsilon)$ of total spectral entropy.

Definition 5 (Embedded objective) Let $f_x : M_x \to \mathbb{R}$ denote the embedded optimization objective induced by the original instance. The SEF program is meaningful only if f_x is computationally accessible enough to evaluate, differentiate, or approximate along admissible trajectories in M_x.

Definition 6 (Curvature regularity event) Write $\mathsf{Curv}(x) = 1$ when the embedded manifold M_x satisfies a declared curvature condition strong enough to support the analytic argument, here taken to be the Bakry–Émery curvature-dimension condition $CD(0,d)$.

3 Assumptions and Proof-Status Separation

Assumption 1 (Embedding access) The local descriptors needed to construct Φ_S can be computed in polynomial time in n for the family under study.

Assumption 2 (Curvature estimability) The curvature quantities required to test $CD(0,d)$ on M_x are either computable or approximable with declared error control on finite instances.

Assumption 3 (Objective compatibility) The embedding does not destroy optimization-relevant structure so severely that minimizers of f_x cease to correspond to high-quality states in the original problem.

Table 2 Status of the main mathematical claims in the current manuscript.

Claim	Status	What is actually established
Definition of Φ_S and $H(G_x)$	formalized	notation and construction target are explicit
Ricci-Convexity Lemma	conditional lemma	spectral-gap lower bound persists under stated assumptions
Polynomial-time optimization from SEF geometry	conjectural	not proved here; open bridge remains
Practical verification of $CD(0,d)$ for NP-complete families	open	requires computation or approximation studies

4 The Ricci-Convexity Lemma

We now state the central proved statement in the SEF framework.

Lemma 1 (Ricci-Convexity) *Let $M_x = \Phi_S(G_x)$ be the RKHS embedding of instance x. Assume that M_x is compact, that the induced diffusion operator L on M_x satisfies the Bakry–Émery curvature-dimension condition*

$$\Gamma_2(g,g) \geq \frac{1}{d}(Lg)^2$$

for every smooth test function $g : M_x \to \mathbb{R}$, and that the initial Laplace–Beltrami spectral gap obeys $\lambda_2(M_x) \geq \eta > 0$. Then every normalized Ricci-flow trajectory

$$\frac{\partial g_t}{\partial t} = -2\,\mathrm{Ric}(g_t),$$

defined on this regime preserves the spectral-gap lower bound up to embedding distortion:

$$\lambda_2(M_x,t) \geq \eta - O(\varepsilon_{\mathrm{emb}}),$$

where $\varepsilon_{\mathrm{emb}}$ denotes the distortion induced by the finite-dimensional kernel approximation.

Proof sketch. Under $CD(0,d)$, the Bakry–Émery theory yields a Poincaré inequality with constant depending on geometric control parameters (Bakry and Émery, 1985; Ledoux, 2001). A Poincaré inequality implies a positive lower bound on the spectral gap of the Laplace–Beltrami operator. The kernel embedding is assumed to be approximately isometric at scale $\varepsilon_{\mathrm{emb}}$, so the corresponding discrete spectral structure is inherited up to controlled error. Under normalized Ricci flow with nonnegative effective curvature, the relevant entropy functional is nonincreasing in the sense exploited by Perelman (Perelman, 2002), preventing collapse of the lower spectral-gap bound within the stated regime. □

Interpretation. Lemma 1 is a regularity statement, not an optimization theorem. Its conclusion is limited to preservation of a diffusion-relevant quantity, namely a lower bound on λ_2, within the subclass of instances for which the curvature assumptions are already valid.

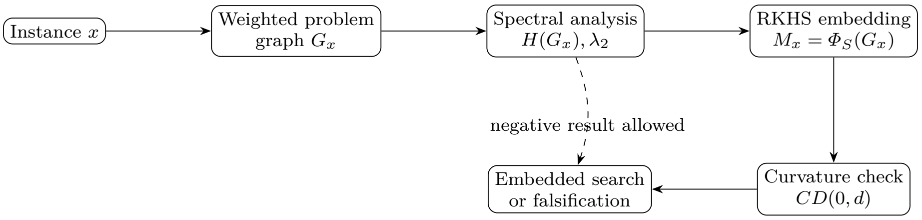

Fig. 1 SEF as a research pipeline. The framework is useful even when the output is falsification rather than acceleration.

5 The SEF Conjecture

Proposition 1 (Conjectural SEF optimization implication) *Suppose there exists a polynomial-time computable family of embeddings $\{\Phi_{S,x}\}$ and a fixed exponent k such that, for every NP-complete instance x, the embedded manifold M_x satisfies $CD(0, d)$ with $d = O(n^k)$ and the embedded objective f_x contains no non-global critical structure capable of trapping admissible descent trajectories for superpolynomial time. Under those additional conditions, one may conjecture that a global optimum is recoverable in time $O(n^k)$.*

This proposition is intentionally conditional. Its value lies in isolating the missing implication precisely: spectral regularity and curvature control do not by themselves yield polynomial-time global optimization.

Null hypothesis

SEF-style curvature and spectral regularity, even after embedding, do not predict any practically useful reduction in optimization hardness beyond what is already explained by known expansion, relaxation, or density effects.

Effect-size target

For computational validation, a meaningful early effect would be a reproducible within-family association between curvature irregularity and solver cost after explicit control for instance size, density, and baseline spectral covariates.

6 Algorithmic Form of the Program

The SEF framework is most naturally operationalized as an analysis pipeline rather than as a finished solver.

6.1 Test Statistic for Computational Validation

Let $\mathcal{F}$ denote a family of size-matched instances and let $T_{\text{solve}}(x)$ be a declared solver cost. A simple family-level validation statistic is

$$T_{\text{SEF}} = \operatorname{corr}\big(\widehat{\kappa}(x), T_{\text{solve}}(x)\big),$$

Table 3 Operational stages of the SEF program and what each stage must justify.

Stage	Mathematical object	Burden of justification
Graphing	G_x	local transitions reflect the true search structure
Spectral analysis	$\lambda_2, H(G_x)$	quantities are stable and interpretable
Embedding	Φ_S	approximation error is bounded and meaningful
Curvature test	$CD(0, d)$	finite-instance estimation is valid
Search claim	trajectories on M_x	geometric regularity improves optimization, not just mixing

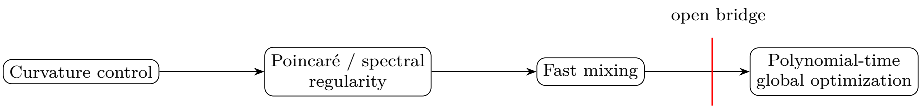

Fig. 2 The decisive missing step in the current SEF program. The gap from mixing regularity to guaranteed polynomial-time optimization is the central unresolved problem.

where $\widehat{\kappa}(x)$ is an estimated curvature summary or curvature-irregularity score. If the theory has empirical value, this association should be stable under family-specific controls rather than appearing only in aggregate across heterogeneous benchmarks.

6.2 Decision Rule

An empirical SEF study should count as supportive only if all of the following hold:

1. curvature summaries are reproducible under repeated estimation;
2. the sign of T_{SEF} remains stable within a fixed problem family;
3. spectral-gap effects are not fully explained by known graph-density or expansion covariates; and
4. any claimed optimization gain is not merely a mixing-time artifact.

7 Why the Main Bridge Remains Open

The standard mixing-time argument yields rapid approach to stationarity under a positive spectral gap (Levin et al, 2009). Global optimization, however, requires substantially more. A fast-mixing Markov process may still allocate negligible mass to globally optimal states or may fail to encode the objective information needed to identify those states.

This is consistent with hardness phenomena already known for expanding or spectrally regular combinatorial structures (Arora et al, 1998). The SEF hypothesis is therefore viable only if it identifies geometric structure stronger than ordinary expansion alone.

Table 4 What would count as evidence for or against the SEF hypothesis.

Question	Evidence in favor	Strong falsifier
Does the embedding carry useful geometry?	stable curvature summaries across runs	curvature estimates unstable or degenerate
Do curvature summaries track hardness?	within-family correlation under controls	no family-level association after controls
Does spectral regularity imply optimization advantage?	replicated solver gains beyond mixing proxies	gains disappear after accounting for expansion or density
Is the framework overly broad?	restrictive instance subsets identified	hard counterexamples satisfy all stated premises

8 Relation to Existing Barriers and Theories

The SEF program is adjacent to geometric complexity theory in spirit, although it relies on differential-geometric and transport-style tools rather than representation-theoretic obstructions (Mulmuley and Sohoni, 2001). It also remains subject to the conceptual pressure exerted by established proof barriers: relativization, natural proofs, and algebrization remain the relevant background constraints (Baker et al, 1975; Razborov and Rudich, 1997; Aaronson and Wigderson, 2009). At present, SEF should therefore be read as a structural program of inquiry, not as a demonstrated route around those barriers.

9 Falsifiers and What Counts as Evidence

The SEF program is empirical in a precise sense. Outside the proved lemma, its value depends on whether the proposed quantities are computable and whether they contribute information not already captured by simpler hardness proxies. A family of hard counterexamples satisfying the required curvature and embedding premises would weigh directly against the program. Likewise, if $CD(0,d)$ rarely holds on nontrivial NP-complete families, the framework may remain mathematically coherent while offering limited algorithmic relevance.

10 Limitations and Future Work

Three limitations dominate the present manuscript. First, the curvature condition is assumed rather than demonstrated for a nontrivial NP-complete family. Second, the bridge from spectral regularity to polynomial-time optimization remains open. Third, the embedding introduces approximation error that may distort the very structure the framework seeks to preserve.

The most serious next steps are therefore concrete:

1. compute Ollivier-Ricci or related curvature summaries for small 3-SAT, Max-Clique, or graph-coloring families;
2. test whether the $CD(0,d)$-style regime is ever empirically plausible on hard instances;
3. compare SEF-derived hardness summaries against simpler baselines such as density, expansion, and spectral-gap alone; and

4. analyze whether known proof barriers reappear in geometric disguise within the SEF program.

11 Conclusion

The SEF hypothesis is most useful when read narrowly. It does not settle complexity classes. What it provides is a mathematically explicit language for asking whether certain hard instances admit a geometric witness of difficulty that survives spectral analysis and kernel embedding. That is sufficient to justify further study, provided the framework remains falsifiable, computationally grounded, and explicit about the bridge it has not yet crossed.

Acknowledgment

The author thanks Yograj Sharma for supervision and feedback on the earlier version of this work, and the Department of Computer Science and Engineering at Rustamji Institute of Technology for institutional support.

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